IRRI
International Rice Research Institute

CGIAR



History of IRRI–JIRCAS Bacterial Blight and Blast Near-Isogenic Lines: Cornerstones of Global Rice Pathology

AUTHORS
Van Schepler-Luu, Mary Jeanie Telebanco-Yanoria, and Casiana M. Vera Cruz

CITATION
Schepler-Luu *et al*., 2025

ACKNOWLEDGEMENTS
The authors would like to acknowledge Dr. Tom W. Mew and Dr. David J. Mackill for their inspiring conversation and for sharing their insights regarding the development of the isogenic lines.

We also extend our gratitude to Dr. Hiroki Saito, Dr. Nobuya Kobayashi and all collaborators from IRRI's Plant Pathology Department, who provided valuable comments and feedback on the significance of the isogenic lines.

1. INTRODUCTION

Near-isogenic lines (NILs) carrying a single resistance gene in a common genetic background have revolutionized rice pathology. By enabling clear, reproducible tests of pathogen virulence and host resistance, these lines are essential for:

- Understanding pathogen populations and their virulence mechanisms
- Assessing synergistic interactions of combined resistance genes
- Identifying effective resistance genes for deployment
- Providing valuable resistance sources for breeding programs worldwide



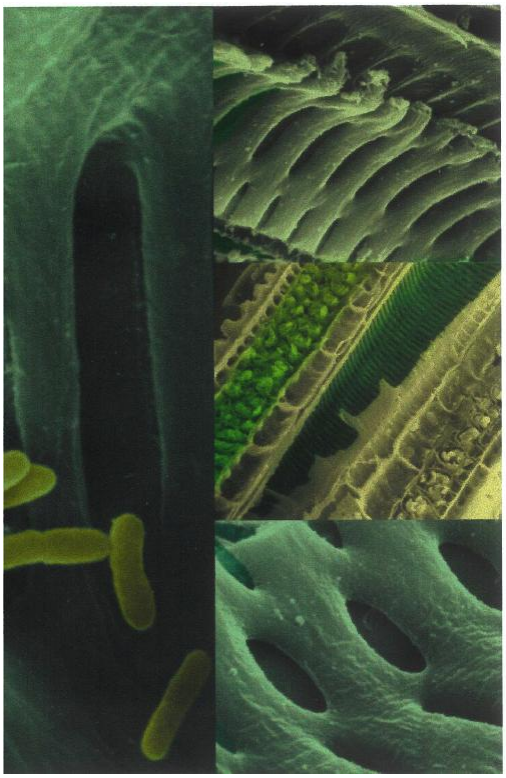
Bacterial blight infection characterized by yellowing and wilting of rice leaves

The IRBB (for bacterial blight) and IRBL (for rice blast) NILs, developed through decades of collaboration between the International Rice Research Institute (IRRI) and the Japan International Research Center for Agricultural Sciences (JIRCAS), have become vital tools in nearly every rice pathology laboratory worldwide. Between 2012 and 2025 alone, the International Rice Genebank distributed over 1,149 samples of these NILs to 26 institutions across 14 countries, underscoring their significant global impact.

2. ORIGINS OF THE BACTERIAL BLIGHT NEAR-ISOGENIC LINES (IRBB)

The concept of bacterial blight near-isogenic lines (NILs) emerged in the late 1970s during the first visit of IRRI Director General Dr. Nyle C. Brady to Japan's JIRCAS (formerly Tropical Agriculture Research Center or TARC). Invited by the Japanese Ministry of Agriculture, Forestry, and Fisheries, Dr. Brady encouraged IRRI scientists to propose collaborative research topics with Japanese counterparts. Among the selected proposals was one from Dr. Tom W. Mew of IRRI's Plant Pathology Department.

Dr. Mew proposed developing bacterial blight (BB) NILs carrying resistance (R) genes identified at IRRI, using the susceptible variety IR8 as the recurrent parent. However, IR8 was later found to carry the *Xa11* resistance gene, prompting the selection of IR24 as the common genetic background for the IRBB NILs.



IRBB10 used to demonstrate vascular defense responses of Xoo (provided by Dr Jan Leach)

With support from Japan's Ministry of Agriculture, Forestry and Fisheries, a five-year collaborative project was launched in the 1980s between IRRI and JIRCAS. This brought together pathologists, breeders, and geneticists from both institutions. Dr. Tsugufumi Ogawa, a geneticist, led NIL development at IRRI's Plant Breeding Department, working closely with IRRI pathologists. Plant Pathologist Dr. Tsuyoshi Yamamoto also joined the project, which was co-led at IRRI by Dr. Mew and Dr. Gurdev S. Khush. Dr. Mew also served as IRRI's Project Leader for the Japan collaboration for over a decade, until a few years before his retirement in 2004. Visiting scientists Dr. Osamu Horino and Dr. Hisatoshi Kaku contributed to the work, which led to the release of the first IRBB lines published in 1989 and 1991 (Ogawa *et al*., 1989; Ogawa *et al*., 1991).

From a plant pathology perspective, this collaboration laid the foundation for early studies on the population variability of *Xanthomonas oryzae* pv. *oryzae* (*Xoo*), the pathogen causing bacterial blight. These efforts, which began in the mid-1970s, confirmed that the strong variety x isolate interaction followed H.H. Flor's gene-for-gene concept in the rice-*Xoo* pathosystem (Mew & Vera Cruz, 1979; Mew *et al*., 1982; Vera Cruz *et al*., 1984; Mew, 1987; Mew *et al*., 1992; Leach *et al*., 1992; Nelson *et al*., 1994; Ardales *et al*., 1996; Vera Cruz *et al*., 1996; Vera Cruz *et al*.,

2000). Researchers identified and characterized differential varieties based on seedling resistance, moderate resistance, or adult plant resistance. Between 1979 and the mid-1980s, in the Philippines, six *Xoo* strains, PXO61 (race 1), PXO86 (race 2), PXO79 (race 3), PXO71 (race 4), PXO112 (race 5), and PXO99 (race 6) were classified to represent the corresponding *Xoo* races 1 to 6. These strains showed specific interactions with a set of differential varieties: IR20 (*Xa4*), IR1545-339-2-2 (*xa5*), Cas 209 (*Xa10*), and DV85 (*xa5*, *Xa7*, *xa24*), with IR8 (*Xa11*) as the susceptible control (Mew & Vera Cruz, 1979; Mew *et al*., 1982; Mew, 1987; Vera Cruz & Mew, 1989; Mew *et al*., 1992).



IR24 and IRBB57 (maturity stage) during BB monitoring in the Philippines

Under IRRI's Germplasm Evaluation and Utilization (GEU) Program, thousands of accessions from the International Rice Genebank were screened. Additional varieties were identified and validated through genetic, molecular, and genomic studies. The original differential set included IR8 (*Xa11*) as the susceptible line; IR20/IR22 (*Xa4*); Cas 209 (*Xa10*); IR1545-339-2-2 (*xa5*); and DV85 (*xa5*, *Xa7*, *xa24*). Later additions included *Xa7* (from DV85), *xa13* (from BJ1), *Xa14* (from TN1), *Xa21* (from *O. longistaminata*), and *Xa23* (from *O. rufipogon*) (Table 1). These differential lines were developed into a set of BB near-isogenic lines (NILs) using IR24 as the recurrent parent (Table 1) (Ogawa & Yamamoto, 1987). These NILs, each carrying individual *Xa* or *xa* genes, have since served as the international differential set for identifying *Xoo* races, guiding breeding programs in Asia, Africa, and beyond (Huang *et al*., 1997; Ogawa *et al*., 1991).

Table 1. Sources of resistance to BB used in developing BB NILs and pyramids.

Line	*R-genes*	Chromosome #	Donor	Accession #	Reference
IRBB1	*Xa1*	4	Kogyoku	66978	Ogawa *et al*., 1978; Yoshimura *et al*., 1998
IRBB3	*Xa3*	11	Wase Aikoku	525	Ogawa *et al*., 1986

IRBB4	*Xa4*	11	TKM6	237	Singh et al., 1983; Ogawa *et al*., 1986, 1988; Yoshimura *et al*., 1995
IRBB5	*xa5*	5	DZ192, IR1545-339-2-2	8518	Sidhu *et al*. 1978, Yoshimura *et al*., 1984; Ogawa and Khush, 1989; Iyer and McCouch, 2004
IRBB7	*Xa7*	6	DV85	8839	Sidhu *et al*., 1978; Lee and Khush 2000
IRBB8	*xa8*	7	PI231129	11114	Sidhu *et al*., 1978
IRBB10	*Xa10*	11	CAS209	15793	Mew *et al*., 1982; Yoshimura *et al*., 1983
IRBB11	*Xa11*	3	IR8, IR944	10320	Kurata and Yamazaki 2006; Goto *et al*., 2009
IRBB13	*xa13*	8	BJ1	3711	Ogawa *et al*., 1988
IRBB14	*Xa14*	4	TN1	105	Sidhu *et al*., 1978; Taura, S. *et al*., 1987
IRBB21	*Xa21*	11	*O. longistaminata*	wild species of *Oryza*	Khush *et al*., 1990; Ronald *et al*., 1992; Song *et al*., 1995
IRBB23	*Xa23*	11	*O. rufipogon*	wild species of *Oryza*	Zhang *et al*., 1998, 2001; Wang *et al*., 2014

Later, *Xoo* was established as the model pathogen for rice (Nino-Liu *et al*., 2006). The ten Philippines' *Xoo* races identified to date have been differentiated using varieties carrying specific *Xa* or *xa* genes for bacterial blight (BB) resistance (Table 2).

To enhance resistance to diverse *Xoo* races globally, various combinations of *Xa/xa* genes conferring different mechanisms of resistance have been pyramided into the IR24 background (Huang *et al*., 1997). Pyramid lines IRBB61 to IRBB66, incorporating various combinations of *Xa4*, *xa5*, *Xa7*, *xa13*, and *Xa21*, were developed in the year 2000 during the final phase of the Asian Rice Biotechnology Network at IRRI. The pyramid lines with effective gene combinations were shared with Asian partners for improving BB resistance in their popular commercial varieties (Leung *et al*., 2004). The effectiveness of each gene combination varied depending on the virulence spectrum of *Xoo* races in different regions (Table 2).

More recently, robust genes such as *Xa23* and *Xa27* have been introgressed into the IR24 background; however, no pyramid lines have yet been developed from these single-gene BB NILs. Newly identified *Xa* genes from wild rice including *Xa48*, *xa45(t)*, *Xa38*, and *Xa33* also show strong potential for pyramiding into IR24 and other suitable genetic backgrounds (Lore *et al*., 2025). These new genes and their future pyramids are expected to provide durable resistance against evolving *Xoo* races.

Table 2. IRBB lines and pyramids developed through IRRI–JIRCAS Collaboration showing specific interactions with differential races of the Philippines *Xanthomonas oryzae* pv. *oryzae*[a].

Line	*R-genes*	Race 1 PXO 61[b]	Race 2 PXO 86	Race 3B PXO 79	Race 3C PXO 340	Race 4 PXO 71	Race 5 PXO 112	Race 6 PXO 99	Race 7 PXO 145	Race 8 PXO 280	Race 9a PXO 339	Race 9b PXO 349	Race 9c PXO 347	Race 9d PXO 363	Race 10 PXO 341
IRBB1	*Xa1*	S	S	S	S	S	S	S	S	S	S	S	S	S	S
IRBB3	*Xa3*	S	S	S	S	S	S	S	S	S	S	S	S	S	S
IRBB4	*Xa4*	R	S	S	S	MS	R	S	R	R	S	S	S	S	R
IRBB5	*xa5*	R	R	R	R	S	R	S	R	R	R	R	R	R	R
IRBB7	*Xa7*	MS	R	R	R	S	R	S	R	R	S	MS	S	MS	MS
IRBB8	*xa8*	S	S	S	S	S	MR	S	S	S	S	S	S	S	S
IRBB10	*Xa10*	S	R	S	S	S	R	S	R	S	S	S	S	S	S
IRBB11	*Xa11*	S	S	S	S	S	S	S	S	S	S	S	S	S	S
IRBB13	*xa13*	S	S	S	S	S	S	R	S	S	S	S	S	S	S
IRBB14	*Xa14*	S	S	S	S	S	R	S	S	R	S	S	S	S	S
IRBB21	*Xa21*	R	R	R	R	R	R	R	R	R	R	R	R	R	S
IRBB23	*Xa23*	R	R	R	R	R	R	R	R	R	R	R	R	R	R
IRBB50	*Xa4 + xa5*	R	R	R	R	R	R	S	R	R	R	R	R	R	R
IRBB51	*Xa4 + xa13*	R	S	S	S	MS	R	R	R	MR	S	-	-	-	R
IRBB52	*Xa4 + Xa21*	R	R	R	MR	MR	R	MR	R	R	MR	-	-	-	R
IRBB53	*xa5 + xa13*	-	-	-	-	-	-	-	-	-	-	-	-	-	-
IRBB54	*xa5 + Xa21*	R	R	R	R	R	R	R	R	R	R	-	-	-	R
IRBB55	*xa13 + Xa21*	R	R	R	MR	R	R	R	MR	MR	MR	-	-	-	S
IRBB56	*Xa4 + xa5 + xa13*	R	R	R	R	R	R	MR	R	R	R	-	-	-	R
IRBB57	*Xa4 + xa5 + Xa21*	R	R	R	R	R	R	R	R	R	R	-	-	-	R
IRBB59	*xa5 + xa13 + Xa21*	R	R	R	R	R	R	R	R	R	R	-	-	-	R
IRBB60	*Xa4 + xa5 + xa13 + Xa21*	R	R	R	R	R	R	R	R	R	R	-	-	-	R
IRBB61	*Xa4 + xa5 + Xa7*	R	R	R	R	MR	R	S	R	R	R	-	-	-	R
IRBB62	*Xa4 + Xa7 + Xa21*	R	R	R	R	R	R	MR	R	R	MS	-	-	-	R

IRBB63	*xa5 + Xa7 + xa13*	R	R	R	R	R	R	R	R	R	R	-	-	-	R
IRBB64	*Xa4 + xa5 + Xa7 + Xa21*	R	R	R	R	R	R	R	R	R	R	-	-	-	R
IRBB65	*Xa4 + Xa7 + xa13 + Xa21*	R	R	R	R	R	R	R	R	R	MR	-	-	-	R
IRBB66	*Xa4 + xa5 + Xa7 + xa13 + Xa21*	R	R	R	R	R	R	R	R	R	R	-	-	-	R
IRBB67	*Xa4 + Xa7*	-	-	-	-	-	-	-	-	-	-	-	-	-	-
IR24	*Xa18 (susceptible recurrent)*	S	S	S	S	S	S	S	S	S	S	S	S	S	S

[a]Resistance or susceptibility of rice plants to *Xanthomonas oryzae* pv. *oryzae* is expressed in lesion lengths measured 14 days post inoculation. Resistant (R): <5 cm; moderately resistant (MR): 5 – 10 cm; moderately susceptible (MS): 10 – 15 cm; susceptible (S): >15 cm. All NILs were inoculated at 40-45 days after sowing.

[b]Representative strain of each race in *Xanthomonas oryzae* pv. *oryzae* in the Philippines.

3. DEVELOPMENT OF NEAR-ISOGENIC LINES FOR RICE BLAST (IRBL)

When the IRBB project was successfully launched, Dr. Mike Bonman, an IRRI plant pathologist specializing in rice blast, initiated the development of blast NILs using CO39 as the genetic background, in collaboration with Dr. David Mackill, an IRRI breeder, a former head of IRRI's Plant Breeding Division. The initial set of blast NILs in the susceptible variety CO39 was published in 1991 (Mackill, 1992).

In 1995, Dr. Tokio Imbe of JIRCAS expanded this work by incorporating a broader range of blast resistance (*Pi*) genes from diverse donor sources into various genetic backgrounds. This enabled more precise differential analyses of *Magnaporthe oryzae* populations.

From 1999 to 2010, Dr. Hiroshi Kato, Dr. Yoshimichi Fukuta, and Dr. Nobuya Kobayashi of JIRCAS continued and led the development of these NILs across different genetic backgrounds, promoting the use of a comprehensive set of monogenic lines in LTH backgrounds and NILs in CO39, LTH, US-2, IR49830-7-1-2-2, and IR64 backgrounds (Table 3 and 4). Their landmark 2000 publication described 24 monogenic lines in LTH (Tsunematsu *et al.*, 2000), establishing the foundation for the International Differential System for Rice Blast.

4. THE ROLE OF THE INTERNATIONAL RICE BLAST RESEARCH NETWORK (IRBRN)

Established in 2003 under the leadership of Dr. Fukuta, the International Research Program for Blast Research Network (IRBRN) aimed to strengthen global collaboration by:

- Evaluating blast resistance using standardized monogenic lines
- Characterizing *Magnaporthe oryzae* isolates across regions
- Enhancing cooperation among international breeding programs

A major milestone was the standardization of the differential system in 2008 using the IRBL lines in the LTH background, which became the global benchmark for blast resistance research (Telebanco-Yanoria *et al*., 2008).

Table 3. International Rice Blast Monogenic/Near Isogenic Lines (IRRI–JIRCAS)

Type	No. of Genes	Resistance Genes	Background	Reference
IL	4	*Pi1*, *Pi3*, *Piz5*, *Pita*	CO39	Mackill *et al*., 1992
ML	24	*Pia*, *Pib*, *Pik*, *Pik-h*, *Pish*, *Pit*, *Pita*, *Pi9*, *etc*.	LTH	Tsunematsu *et al*., 2000
NIL	11	*Pib*, *Piz-5*, *Pi9*, *Pi3*, *Pia*, *Pik-s*, *Pik*, *Pik-h*, *Pi7(t)*, *Pita*, *Pita-2*	LTH	Telebanco-Yanoria *et al*., 2010
NIL	14	*Pish*, *Pib*, *Piz-5*, *Piz-t*, *Pi5(t)*, *Pik-s*, *Pik*, *Pik-h*, *Pik-m*, *Pik-p*, *Pi1*, *Pi7(t)*, *Pita*, *Pita-2*	CO39	Telebanco-Yanoria *et al*., 2010
NIL	18	*Pia*, *Pib*, *Pik*, *Pik-h*, *Pish*, *Pit*, *Pita*, *Pi9*, *etc*	US-2	Fukuta *et al*., 2022
NIL	8	*Pia*, *Pi3*, *Pi5*, *Pi7*, *Pi9*, *Piz*, *Pish*, *Pita-2*	IR49830-7-1-22	Koide *et al*., 2011
NIL	11	*Pish*, *Piz5*, *Piz*, *Pi9*, *Pii*, *Pi3*, *Pik*, *Pikp*, *Pikh*, *Pi1*, *Pi7*	IR64	Fukuta *et al*., 2022

These lines are widely used for global blast pathotyping and breeding.

Table 4. Detailed list of all available Blast Monogenic/Near Isogenic lines (provided by Dr. Fukuta)

Blast Resistant Monogenic Lines (24 genes)

Sl. No.	Designation in recent publications	Seed source (JIRCAS no.)	Rice blast resistant genes in recent publications
1	LTH	BMsf-1	*Susceptible recurrent*
2	IRBLsh-B	BMsf-2	*Pish*
3	IRBLsh-S	BMsf-3	*Pish*
4	IRBLb-B	BMsf-4	*Pib*
5	IRBLt-K59	BMsf-5	*Pit*
6	IRBLa-A	BMsf-6	*Pia*
7	IRBLa-C	BMsf-7	*Pia*
8	IRBLi-F5	BMsf-8	*Pii*
9	IRBL3-CP4	BMsf-9	*Pi3*
10	IRBL5-M	BMsf-10	*Pi5(t)*
11	IRBLks-F5	BMsf-11	*Pik-s*
12	IRBLks-S	BMsf-12	*Pik-s*
13	IRBLkm-Ts	BMsf-13	*Pik-m*
14	IRBL1-CL	BMsf-14	*Pi1*
15	IRBLkh-K3	BMsf-15	*Pik-h*
16	IRBLk-Ka	BMsf-16	*Pik*
17	IRBLkp-K60	BMsf-17	*Pik-p*
18	IRBL7-M	BMsf-18	*Pi7(t)*
19	IRBL9-W	BMsf-19	*Pi9*
20	IRBLz-Fu	BMsf-20	*Piz*
21	IRBLz5-CA-1	BMsf-21	*Piz-5/ Pi2*(t)
22	IRBLzt-T	BMsf-22	*Piz-t*
23	IRBLta2-Pi	BMsf-23	*Pita-2*
24	IRBLta2-Re	BMsf-24	*Pita-2*
25	IRBL12-M	BMsf-25	*Pi12(t)*
26	IRBLta-CP1	BMsf-26	*Pita*
27	IRBLta-K1	BMsf-27	*Pita*
28	IRBLta-CT2	BMsf-28	*Pita*
29	IRBL19-A	BMsf-29	*Pi19(t)*
30	IRBL20-IR24	BMsf-30	*Pi20(t)*
31	IRBL11-Zh	BMsf-31	*Pi11(t)*

Blast Resistant LTH NILs (11 genes)

32	IRBLb-B[LT]	BMsf-32	*Pib*
33	IRBLz5-CA[LT]	BMsf-33	*Piz-5/Pi2*(t)
34	IRBL9-W[LT]	BMsf-34	*Pi9(t)*
35	IRBL3-CP4[LT]	BMsf-35	*Pi3*
36	IRBLa-Ze[LT]	BMsf-36	*Pia*
37	IRBLks-S[LT]	BMsf-37	*Pik-s*
38	IRBLKs-B40[LT]	BMsf-38	*Pik-s*
39	IRBLKs-Zh[LT]	BMsf-39	*Pik-s*
40	IRBLk-Ka[LT]	BMsf-40	*Pik*
41	IRBLkh-K3[LT]	BMsf-41	*Pik-h*
42	IRBL7-M[LT]	BMsf-42	*Pi7(t)*
43	IRBLk*-Du[LT]	BMsf-43	*Pik**
44	IRBLk*-NP[LT]	BMsf-44	*Pik**
45	IRBLk*-F14[LT]	BMsf-45	*Pik**
46	IRBLk*-F25[LT]	BMsf-46	*Pik**
47	IRBLk*-F66[LT]	BMsf-47	*Pik**
48	IRBLta-CT2[LT]	BMsf-48	*Pita/Pi4*(t)
49	IRBLta-K1[LT]	BMsf-49	*Pita/Pi4(t)*
50	IRBLta-Zh[LT]	BMsf-50	*Pita/ Pi4(t)*
51	IRBLta2-Pi[LT]	BMsf-51	*Pita-2*

Blast Resistant CO39 NILs (14 genes)

52	CO39	BMsf-61	*Pia*
53	IRBLsh-Ku[CO]	BMsf-62	*Pish*
54	IRBLsh-S[CO]-1	BMsf-63	*Pish*
55	IRBLsh-B[CO]	BMsf-64	*Pish*
56	IRBLsh-Fu[CO]	BMsf-65	*Pish*
57	IRBLb-IT13[CO]	BMsf-66	*Pib*
58	IRBLz5-CA[CO]	BMsf-68	*Piz5*
59	IRBL5-M[CO]	BMsf-70	*Pi5(t)*
60	IRBLks-CO[CO]	BMsf-71	*Pik-s*
61	IRBLk-Ku[CO]	BMsf-72	*Pik*
62	IRBLk-Ka[CO]	BMsf-73	*Pik*
63	IRBLkh-K3[CO]	BMsf-74	*Pik-h*
64	IRBLkm-Ts[CO]	BMsf-75	*Pik-m*
65	IRBLkp-K60[CO]	BMsf-76	*Pik-p*
66	IRBLk*-F21[CO]	BMsf-77	*Pik**
67	IRBLk*-F14[CO]	BMsf-78	*Pik**
68	IRBLk*-F25[CO]	BMsf-79	*Pik**
69	IRBLk*-F40[CO]	BMsf-80	*Pik**
70	IRBLk*-F66[CO]	BMsf-82	*Pik**
71	IRBLk*-KU86[CO]	BMsf-83	*Pik**

72	IRBLta-Ya[CO]	BMsf-86	*Pita/ Pi4(t)*
73	IRBLta-Me[CO]	BMsf-87	*Pita/ Pi4(t)*
74	IRBLta2-Pi[CO]	BMsf-88	*Pita-2*
75	IRBLta2-Re[CO]	BMsf-89	*Pita-2*
76	IRBLta2-IR64[CO]	BMsf-90	*Pita-2*
77	IRBL1-LA[CO]	BMsf-91	*Pi1*
78	IRBL1-CL[CO]	BMsf-92	*Pi1*

Blast Resistant US2 Lines (18 genes)

79	US-2	BMsf-93	
80	US2NILPia-Aichiasahi*	BMsf-101	*Pia*
81	US2NILPia-C	BMsf-102	*Pia*
82	US2NILPii-F5*	BMsf-98	*Pii*
83	US2NILPi3-CP4*	BMsf-99	*Pi3*
84	US2NILPi5-M*	BMsf-100	*Pi5(t)*
85	US2NILPiks-F5	BMsf-103	*Pik-s*
86	US2NILPiks-S*	BMsf-104	*Pik-s*
87	US2NILPikm-TS*	BMsf-107	*Pik-m*
88	US2NILPi1-CL*	BMsf-106	*Pi1*
89	US2NILPikh-K3*	BMsf-105	*Pik-h*
90	US2NILPik-Ka*	BMsf-108	*Pik*
91	US2NILPikp-K60*	BMsf-109	*Pik-p*
92	US2NILPi7(t)-M*	BMsf-110	*Pi7(t)*
93	US2NILPi9-W*	BMsf-94	*Pi9*
94	US2NILPiz-Fu*	BMsf-95	*Piz*
95	US2NILPiz5-CA*	BMsf-96	*Piz-5/ Pi2(t)*
96	US2NILPizt-T*	BMsf-97	*Piz-t*
97	US2NILPita2-Re*	BMsf-111	*Pita-2*
98	US2NILPita-K1*	BMsf-112	*Pita/Pi4(ta)*
99	US2NILPita-CP1*	BMsf-113	*Pita/Pi4(ta)*
100	US2NILPita-CT2*	BMsf-114	*Pita/Pi4(ta)*
101	US2NILPi20-IR24*	BMsf-115	*Pi20(t)*

Blast Resistant IR49830-7-1-2-2 NILs (9 genes)

102	IR49830-7-1-2-2	BMsf-118	*Pia, Pib, Pik-s, Pita, and Pi11*(t)
103	IRBL9-W[RL]	BMsf-119	*Pi9*
104	IRBLz5-CA[RL]	BMsf-120	*Piz5*
105	IRBL3-CP4[RL]	BMsf-121	*Pi3*
106	IRBL5-M[RL]	BMsf-122	*Pi5*
107	IRBL7-M[RL]	BMsf-123	*Pi7*
108	IRBLk-Ku[RL]	BMsf-124	*Pik*
109	IRBLsh-Fu[RL]	BMsf-125	*Pish*
110	IRBLta2-Pi[RL]	BMsf-126	*Pita2*
111	IRBLsh-T[RL]	BMsf-127	*Pish*

Blast Resistant IR64 NILs (11 genes)

112	IR 64	BMsf-128	*Pia*, *Pik-s*, *Pib*, *Piz-t*, *Pi20*(t) *or Pita + unknown*
113	IR64NILPik-Ka	BMsf-129	*Pik*
114	IR64NILPik-Ka(ML)	BMsf-130	*Pik*
115	IR64NILPik-Ka(CO)	BMsf-131	*Pik*
116	IR64NILPikh-K3	BMsf-132	*Pikh*
117	IR64NILPikh-K3(CO)	BMsf-133	*Pikh*
118	IR64NILPikp-K60	BMsf-134	*Pikp*
119	IR64NILPikp-K60(CO)	BMsf-135	*Pikp*
120	IR64NILPi1-CL(ML)	BMsf-136	*Pi1*
121	IR64NILPi1-CL(CO)	BMsf-137	*Pi1*
122	IR64NILPi7-M(ML)	BMsf-138	*Pi7(t)*
123	IR64NILPi7-M(CO)	BMsf-139	*Pi7(t)*
124	IR64NILPiz-Fu(ML)	BMsf-140	*Piz*
125	IR64NILPiz5-CA(ML)	BMsf-141	*Piz-5*
126	IR64NILPiz5-CA(CO)	BMsf-142	*Piz-5*
127	IR64NILPi9-W	BMsf-143	*Pi9*
128	IR64NILPi9-W(ML)	BMsf-144	*Pi9*
129	IR64NILPish-S(CO)	BMsf-146	*Pish*
130	IR64NILPish-Ku(CO)	BMsf-147	*Pish*
131	IR64NILPish-S(ML)	BMsf-148	*Pish*
132	IR64NILPii-F5(ML)	BMsf-149	*Pii*
133	IR64NILPi3-CP4(ML)	BMsf-150	*Pi3*

5. THE GLOBAL LEGACY OF IRBB AND IRBL LINES: WHAT PARTNERS SAY



The near-isogenic lines (NILs) for bacterial blight (IRBB) and rice blast (IRBL) developed at IRRI have become global public goods, enabling researchers and breeders across continents to better understand pathogen populations, deploy durable resistance, and improve food security in rice-growing regions.

As *Xanthomonas* and *Magnaporthe* populations continue to evolve, and with climate change further complicating disease dynamics, these shared resources remain as relevant today as when they were first developed. Below, partners from around the world share how IRBB and IRBL lines have advanced research, breeding, and impact on the ground:

“The near-isogenic IRBB lines have been instrumental in advancing our understanding of single-gene resistance in rice. These lines not only enabled the cloning of multiple bacterial blight resistance genes, but also facilitated the identification, cloning, and functional analysis of corresponding effector genes. Their continued use today underscores their enduring value to the research community. The development of these lines reflects the remarkable foresight of their creators.”
— Dr. Jan Leach, Colorado State University, USA

“The IRBB and IRBL lines are a model of international collaboration. They have given us a common language to understand and manage rice diseases across continents. Our partnership with IRRI has helped ensure that breeders and pathologists worldwide can deploy effective

resistance strategies tailored to their local pathogens."
— Dr. Hiroki Saito, Japan International Research Center for Agricultural Sciences (JIRCAS), Japan

"IRBB and IRBL lines have been extensively used in India for breeding varieties resistant to bacterial blight and blast, as well as for pathotyping and profiling pathogen races across the country. Some of the resistant varieties developed using IRBB lines include Improved Pusa Basmati-1, Improved Samba Mahsuri, DRR Dhan 58, and DRR Dhan 60. DRR Dhan 62, and Pusa Samba were developed using IRBL lines. These varieties are widely cultivated across the country. The real impact of IRBB and IRBL lines is felt in the fields of smallholder farmers-reducing crop loss and pesticide use."
— Dr. Raman Sundaram, ICAR-Indian Institute of Rice Research (ICAR-IIRR), India"

At BRRI, we used IRBB60 as the donor and BRRI Dhan29 as the recipient to develop the bacterial blight-resistant variety BRRI dhan101, released in 2021 for the *Boro* season. It is now grown in farmers' fields. IRBB60 was also crossed with BRRI Dhan28 to develop two advanced lines with strong resistance in both artificial and hotspot screenings. Genotyping confirmed the presence of *Xa4*, *xa13*, and *Xa21*. Additional crosses have produced three advanced resistant lines containing *Xa4* and *Xa21*, which are now used as donors in our breeding program."
— Dr. Shireen Quazi, Bangladesh Rice Research Institute (BRRI), Bangladesh

"In Indonesia, the IRBB lines have greatly contributed to the development of high-yielding rice varieties resistant to bacterial blight. The most recent widely cultivated variety, Inpari 32, released in 2013 from a Ciherang × IRBB64 cross, is resistant to BB pathotype III and moderately resistant to pathotypes IV and VIII."
— Dr. Nafisah Nafisah, Indonesian Center for Rice Instrument Standard Testing (ICRIST), Indonesia

"In Madagascar, IRBB lines were used to evaluate local strains of *Xanthomonas oryzae* pv. *oryzae*. Crosses between widely grown cultivars and resistant IRBB lines are now underway to develop new varieties with durable resistance."
— Dr. Harinjaka Raveloson, FOFIFA, Centre Régional de Recherches, Madagascar

"As a rice pathologist, I use these NILs to track the emergence of new virulent races and guide breeding decisions that protect farmers' yields. This is science that directly benefits people on the ground."
— Dr. Jennifer Niones, Philippine Rice Research Institute (PhilRice), Philippines
"Near-isogenic lines (NILs) for bacterial blight (IRBB) and rice blast (IRBL) developed by IRRI scientists has been instrumental in dissecting pathogen population dynamics and breeding durable resistance. For us, these NILs served as differential panels to identify changing virulence profiles of *Xanthomonas oryzae* pv. *oryzae* and *Magnaporthe oryzae* populations in India, and accordingly device pathogen-informed breeding strategies for broad-spectrum and durable resistance. By tracking shifts in pathogen population that could defeat specific resistance-genes,

we are able to anticipate resistance erosion in the field and design effective gene-pyramided rice hybrids accordingly. Overall, these NIL panels helped directly to inform the breeding of more durable, multi-gene resistance in elite backgrounds, thereby accelerating marker-assisted introgression and pyramiding of effective combinations tailored to evolving pathogen populations in across target countries.
— Dr. Deo Mishra, Asia Breeding – Plant Health, Bayer CropScience, India

"IRRI's differential lines have allowed us to map the effectiveness of resistance genes against local pathogen populations with greater accuracy. This regional insight is critical for developing durable disease resistance in commercial rice hybrids."
— Dr. Raman Babu, Distinguished Laureate & South Asia Seed Product Development Lead, Corteva Agriscience, India

6. FUTURE PERSPECTIVES

As climate change intensifies disease pressure and accelerates the evolution of pathogen populations, the importance of NILs in rice research and breeding will only continue to grow. To sustain and expand their impact, future efforts will focus on five key areas:

- **Expanding NIL collections** to incorporate novel resistance genes identified through advanced genomics, genome-wide association studies (GWAS), and exploration of wild rice relatives.
- **Pyramiding multiple resistance genes** to enhance the durability of resistance and slow down pathogen adaptation, ensuring long-term effectiveness in farmers' fields.
- **Integrating marker-assisted and genomic selection** into breeding pipelines to accelerate the deployment of NIL-derived insights and resistance traits in elite varieties.
- **Developing NILs for additional stresses**, including sheath blight, tungro, rice false smut, brown spot to address the broader spectrum of constraints faced by rice farmers.
- **Strengthening international networks** such as the International Rice Blast Research Network (IRBRN) and supporting new pathogen monitoring systems to promote data sharing, harmonize evaluation protocols, and build global capacity in disease surveillance.

The success of the IRBB and IRBL lines stands as a compelling testament to the power of global scientific collaboration. These resources, and the partnerships behind them, have helped secure food systems in the past and will continue to play a critical role in safeguarding rice production amid emerging challenges.

REFERENCES

Ardales, E. Y., Leung, H., Vera Cruz, C. M., Mew, T. W., Leach, J. E., & Nelson R. J. (1996). Hierarchical analysis of spatial variation of the rice bacterial blight pathogen across diverse agroecosystems in the Philippines. *Phytopathology 86*:241-252.

Fukuta, Y., Telebanco-Yanoria, M. J., Koide, Y., Saito, H., Kobayashi, N., Obara, M., & Yanagihara, S. (2022). Near-isogenic lines for resistance to blast disease, in the genetic background of the Indica Group rice (*Oryza sativa* L.) cultivar IR64. *Field Crops Research*, *282*, 108506.

Goto T., Matsumoto T., Furuya N., Tsuchiya K., & Yoshimura A. (2009). Mapping of bacterial blight resistance gene *Xa11* on rice chromosome 3. *JARQ 43*(3):221-225.

Huang, N., Angeles, E. R., Domingo, J., Magpantay, G., Singh, S., Zhang, G., Kumaravadivel, N., Bennett, J., & Khush, G. S. (1997). Pyramiding of bacterial blight resistance genes in rice: marker-assisted selection using RFLP and PCR. *Theor. Appl. Genet.*, *95*(3),313–320.

Iyer-Pascuzzi, A. S. & McCouch, S. R. (2007). Recessive resistance genes and the *Oryza sativa*-*Xanthomonas oryzae* pv. *oryzae* pathosystem. *Mol. Plant Microbe Interact*. *20*,731–739.

Khush, G. S, Bacalangco E., & Ogawa T. (1990). A new gene for resistance to bacterial blight from *O. longistaminata*. *Rice Genet Newsl* 7:121-122.

Koide, Y., Ebron, L. A., Kato, H., Tsunematsu, H., Telebanco-Yanoria, M. J., Kobayashi, N., Yokoo, M., Maruyama, S., Imbe, T., & Fukuta, Y. (2011). A set of near-isogenic lines for blast resistance genes with an Indica-type rainfed lowland elite rice (*Oryza sativa* L.) genetic background. *Field Crops Research*, *123*(1), 19–27.

Kurata, N. & Yamazaki, Y. (2006). Oryzabase. An integrated biological and genome information database for rice. *Plant Physiol*. *140*,12–17.

Leach, J. E., Rhoads M. L, Vera Cruz C. M., White F. F, Mew T. W., & Leung H. (1992). Assessment of genetic diversity and population structure of *Xanthomonas oryzae* pv. *oryzae* using a repetitive DNA element. Appl. Environ. *Microbiol*. *58*:2188-2195.

Lee, K. S. & Khush, G. S. (2000). Genetic analysis of resistance to bacterial blight, *Xanthomonas oryzae* pv. *oryzae* rice. *Rice Genet*. *Newsl*. 17,72.

Leung, H., Wu, J., Liu, B., Bustamam, M., Sridhar, R., Singh, K., Redona, E. D., Quang, V. D., Zheng, K., Bernardo, M., Wang, G., Leach, J., Choi, I. R., & Vera Cruz, C. M. (2004). Sustainable disease resistance in rice: current and future strategies. In: Fischer, T., Turner, N., Angus, J., et al., (Eds.), Proc. 4th International Crop Science Congress. Brisbane, Australia, 26 September–1 October (Published on CD-ROM).

Lore, J. S., Pinili D., Bhatia D., Balloyan L. R., Navea I.P., Hunjan M. S., Mangat G.S., Singh R., Platten J. D., Yang B., Leach, J. E., White, F. F., Vera Cruz, C.M., & Schepler-Luu, V. (2025). Pathogen-informed strategies for durable resistance in rice: Lessons from bacterial blight. *Annu. Rev. Phytopathol*. Vol. 63.

Mackill, D. J. (1992). Inheritance of Blast Resistance in Near-Isogenic Lines of Rice. *Phytopathology*, *82*(7),746.

Mew, T. W. & Vera Cruz C. M. (1979). Variability of *Xanthomonas oryzae*: Specificity in infection of rice differentials. *Phytopathology 69*(2):152-155.

Mew, T. W., Vera Cruz, C. M., & Reyes, R. C. (1982). Interaction of *Xanthomonas campestris* pv. *oryzae* and a Resistant Rice Cultivar. *Phytopathology*, *72*(7),786–786. https://doi.org/10.1094/phyto-72-786

Mew T. W. (1987). Current status and future prospects of research on bacterial blight of rice. *Annu. Rev. Phytopathol. 25*:359-382.

Mew, T. W., Vera Cruz, C. M., & Medalla, E. S. (1992). Changes in race frequency of *Xanthomonas oryzae* pv. *oryzae* in response to rice cultivars planted in the Philippines. *Plant Dis. 76*:1029-1032.

Nelson, R. J., Baraoidan, M. R., Cruz, C. M. V., Yap, I. V., Leach, J. E., Mew, T. W., & Leung, H. (1994). Relationship between Phylogeny and Pathotype for the Bacterial Blight Pathogen of Rice. *Applied and Environmental Microbiology*, *60*(9), 3275–3283. https://doi.org/10.1128/aem.60.9.3275-3283.1994

Nino-Liu, D. O., Ronald, P. C., & Bogdanove, A. J. (2006). *Xanthomonas oryzae* pathovars: model pathogens of a model crop. *Molecular Plant Pathology*, *7*(5), 303–324. https://doi.org/10.1111/j.1364-3703.2006.00344

Ogawa T., Morinaka T., Fujii K., & Kimura T. (1978). Inheritance of resistance of rice varieties of Kogyoku and Java 14 to bacterial group V of *Xanthomonas oryzae*. *Ann. Phytopathol. Soc. Jpn. 44*: 137-141.

Ogawa T., Tabien R. E., Busto G. A., & Khush G. S., Mew T. W. (1986). The relationship of *Xa-3*, *Xa-4*, and *Xa-4b* for resistance to rice bacterial blight. *Rice Genet. Newsl*. 3:83-84.

Ogawa T., Yamamoto T., Khush G. S., & Mew T. W. (1986). The *Xa-3* gene for resistance to Philippine races of bacterial blight of rice, *Rice Genet. Newsl.* 3:77-78.

Ogawa, T., & Yamamoto, T. (1987). Selection of recurrent parents to develop near-isogenic lines resistant to bacterial blight of rice. *JARQ*, 21, 65–69.

Ogawa, T. (1987). Gene symbols for resistance to bacterial blight. *Rice Genet. Newsl.*, 4, 41–43.

Ogawa, T., Yamamoto, T., Khush, G. S., Mew, T. W., & Kaku, H. (1988). Near-isogenic lines as international differentials for resistance to bacterial blight of rice. *Rice Genet. Newsl*. 5, 106–109.

Ogawa T. & Khush G. S. (1989). Major genes for resistance to bacterial blight in rice. Pages 177-192. *In* Bacterial Blight of Rice. IRRI, Los Banos, Philippines.

Ogawa, T., Yamamoto, K., Khush, G. S., & Mew, T. W. (1991). Breeding of near-isogenic lines of rice with single genes for resistance to bacterial blight pathogen (*Xanthomonas campestris* pv. *oryzae*). *Japanese Journal of Breeding, 41*(3), 523–529.

Ronald, P. C., Albano, B., Tabien, R., Abenes, L., Wu, K., McCouch, S., & Tanksley, S. D. (1992). Genetic and physical analysis of the rice bacterial blight disease resistance locus, *Xa21*. *Molecular and General Genetics*, *236*(1), 113–120.

Sidhu, G. S., Khush, G. S., & Mew, T. W. (1978). Genetic analysis of bacterial blight resistance in seventy-four cultivars of rice, *Oryza sativa* L. *Theor. Appl. Genet*, *53*(3), 105–111.

Singh, R. J., Khush, G. S., & Mew, T. W. (1983). A new gene for resistance to bacterial blight in rice. *Crop Science*, *23*(3), 558–560.

Singh, R. J., Khush, G. S., & Mew, T. W. (1983). A new gene for resistance to bacterial blight in rice. *Crop Scien*ce, 23, 558–560.

Song, W.-Y., Wang, G. L., Chen, L.L., Kim, H. S., Pi, L. Y. ., Holsten, T., Gardner, J., Wang, B., Zhai, W.-X. ., Zhu, L. H., Fauquet, C., & Ronald, P. (1995). A Receptor Kinase-Like Protein Encoded by the Rice Disease Resistance Gene, Xa21. *Science*, 270 (5243), 1804–1806.

Taura, S., Ogawa T., Tabien R. E, Khush G. S., Yoshimura A., & Omura T. (1987). The specific reaction of Taichung Native I to Philippine race of bacterial blight and inheritance of resistance to race 5 (PXOI12). *Rice Genet. Newsl.* 4:101-102.

Telebanco-Yanoria, M. J., Imbe, T., Kato, H., Tsunematsu, H., Ebron, L. A., Cruz, C. M. V., Kobayashi, N., & Fukuta, Y. (2008). A Set of Standard Differential Blast Isolates (*Magnaporthe grisea* (Hebert) Barr.) from the Philippines for Rice (*Oryza sativa* L.) Resistance. *Japan Agricultural Research Quarterly JARQ*, *42*(1), 23–34.

Telebanco-Yanoria, M. J., Koide, Y., Fukuta, Y., Imbe, T., Kato, H., Tsunematsu, H., & Kobayashi, N. (2010). Development of near-isogenic lines of Japonica-type rice variety Lijiangxintuanheigu as differentials for blast resistance. *Breeding Science*, *60*(5), 629–638.

Telebanco-Yanoria, M. J., Koide, Y., Fukuta, Y., Imbe, T., Tsunematsu, H., Kato, H., Ebron, L. A., Nguyen, T. M. N., & Kobayashi, N. (2010). A set of near-isogenic lines of Indica-type rice variety CO39 as differential varieties for blast resistance. *Molecular Breeding*, *27*(3), 357–373.

Tsunematsu, H., Yanoria, M. J. T., Ebron, L. A., Hayashi, N., Ando, I., Kato, H., Imbe, T., & Khush, G. S. (2000). Development of monogenic lines of rice for blast resistance. *Breeding Science*, *50*(3), 229–234.

Vera Cruz, C. M., Ardales, E. Y., Skinner, D. Z., Talag, J., Nelson, R. J., Louws, F. J., Leung, H., Mew, T. W., & Leach, J. E. (1996). Measurement of haplotypic variation in *Xanthomonas oryzae* pv. *oryzae* within a single field by Rep-PCR and RFLP analyses. *Phytopathology*, *86* (12), 1352–1359.

Vera Cruz, C. M., Bai, J., Oña, I., Leung, H., Nelson, R. J., Mew, T. W., & Leach, J. E. (2000). Predicting durability of a disease resistance gene based on an assessment of the fitness loss and epidemiological consequences of avirulence gene mutation. *Proceedings of the National Academy of Sciences*, *97*(25), 13500–13505.

Vera Cruz, C. M., Gossele, F., Kersters, K., Segers, P., Van den Mooter, M., Swings, J., & De Ley, J. (1984). Differentiation between *Xanthomonas campestris* pv. *oryzae*, *Xanthomonas campestris*

pv. *oryzicola* and the bacterial brown blotch pathogen by numerical analysis of phenotypic features and protein gel electrophoregrams. *Journal of General Microbiology*, *130*(11), 2983–2999.

Vera Cruz C. M. & Mew T. W. (1989). How variable is *Xanthomonas campestris* pv. *oryzae*? Pages 153-166. *In* Bacterial Blight of Rice. IRRI, Los Banos, Philippines.

Vera Cruz, C. M., Ardales E. Y., Skinner D. Z., Talag, J., Nelson, R. J., Louws F. J., Leung, H. K., Mew T. W., & Leach J. E. (1996). Measurement of haplotypic variation in *Xanthomonas oryzae* pv. *oryzae* within a single field by rep-PCR and RFLP analyses. *Phytopathology*, *86*(12), 1352–1359.

Vera Cruz, C. M., Bai, J., Oña, I., Leung, H., Nelson, R. J., Mew T. W., & Leach, J. E. (2000). Predicting durability of a disease resistance gene based on an assessment of the fitness loss and epidemiological consequences of avirulence gene mutation. Proc. Natl. Acad. Sci. USA 97:13500-13505.

Wang, C., Zhang, X., Fan, Y., Gao, Y., Zhu, Q., Zheng, C., Qin, T., Li, Y., Che, J., Zhang, M., Yang, B., Liu, Y., & Zhao, K. (2014). XA23 is an executor R protein and confers broad-spectrum disease resistance in rice. *Molecular Plant*, *7*(12), 1763–1766.

Yoshimura, A., Mew, T. W., Khush, G. S., & Omura, T. (1983). Inheritance of resistance to bacterial blight in rice cultivar Cas 209. *Phytopathology*, *73*(11), 1409–1412.

Yoshimura, A., Mew, T. W., Khush, G. S., & Omura, T. (1984). Genetics of bacterial blight resistance in a breeding line of rice. *Phytopathology*, *74*(7), 773–777.

Yoshimura, S., Yoshimura, A., Iwata, N., McCouch, S., Abenes, M., Baraoidan, M., Mew, T. W., & Nelson, R. J. (1995). Tagging and combining bacterial blight resistance genes in rice using RAPD and RFLP markers. *Molecular Breeding*, *1*(4), 375–387. https://doi.org/10.1007/bf01248415

Yoshimura, S., Yamanouchi, U., Katayose, Y., Toki, S., Wang, Z. X., Kono, I., Kurata, N., Yano, M., Iwata, N., & Sasaki, T. (1998). Expression of *Xa1*, a bacterial blight-resistance gene in rice, is induced by bacterial inoculation. Proc. Natl. Acad. Sci. U.S.A. 95 (4), 1663–1668.

Zhang, Q., Lin, S. C., Zhao, B. Y., Wang, C. L., Yang, W. C., Zhou, Y. L., Li, D. Y., Chen, C. B., & Zhu, L. H. (1998). Identification and tagging a new gene for resistance to bacterial blight (*Xanthomonas oryzae* pv. *oryzae*) from *O. rufipogon*. *Rice Genetics Newsletter*, *15*, 138–142.

Zhang, Q., Wang, C., Zhao, K., Zhao, Y., Casiana, V. C., Zhu, X. D., & Jiang, L. (2001). The effectiveness of advanced rice lines with new resistance gene *Xa23* to rice bacterial blight. *Rice Genet. Newsl.*, *18*, 71.